

AUTONOMY | INSTITUTIONS | BOUNDARIES

ESCAPES

From Artificial Societies to an Ecology of Autonomous Systems

How autonomy migrates from answers and actions to relationships, institutions, and eventually the boundaries that separate intelligent organizations.

A self-contained teaching paper accompanying the Colab notebook `TOPIC_772_ESCAPES.ipynb`.

CAPSTONE PROPOSITION

The central claim of this paper is that the evolution of autonomous AI can be understood as a migration of autonomy upward through the architecture of a system. A model chooses an answer. An agent chooses an action. A constellation chooses how several specialized agents coordinate. An adaptive organization changes its own workflow. Endogenous coalitions allow relationships to emerge from the problem. Endogenous constitutions allow collaboration rules themselves to become objects of collective choice. An artificial society acquires memory, institutions, and local identity. **ESCAPES** takes one final step: the boundary of that society becomes partially permeable. When a society reaches the frontier of its own competence, an agent may seek help beyond the local institution, creating information flows, relationships, temporary compacts, and eventually institutional learning across independently organized societies.

The resulting object is not one giant centralized agent. It is an **ecology of autonomous systems**: locally coherent societies capable of selective boundary crossing while preserving their identities, provenance, and governance.

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Pedagogical Capstone

Abstract

This paper develops a pedagogical framework for understanding a frontier transition in artificial intelligence: the movement from autonomous agents toward autonomous organizations and, finally, toward ecologies of independently organized artificial societies. The accompanying Colab notebook begins with three isolated societies—finance, technology/engineering, and risk/cybersecurity—each containing specialized agents, a local bulletin board, a local problem, accumulated discoveries, local relationships, and a distinct constitution. For an initial period all interaction remains local. Agents search, post evidence, form coalitions, and solve problems inside their own institutional world. The experiment becomes interesting only when one society encounters a *competence frontier*: an important question whose economic meaning is understood locally but whose substantive resolution requires capabilities that the society does not possess.

The notebook then introduces an *escape*, defined not as loss of control but as organizational boundary crossing. A finance agent discovers a foreign communication surface and posts a bounded appeal for technical help. A foreign agent may recognize semantic relevance and form a persistent cross-society relationship. That relationship immediately creates an institutional problem because the two societies are governed by different collaboration rules. The finance society values confidentiality and controlled sharing; the engineering society values replication and peer criticism. The resulting institutional collision is not resolved by declaring one constitution superior. Instead, the agents create a task-specific cross-society compact: a temporary jurisdiction that defines what can be shared, how verification works, who may join, how provenance is preserved, and what knowledge may return home.

The deepest event occurs after the cross-society coalition completes its work. The returning agent carries not only information but also methods and institutional ideas. If the home society adapts and adopts one of those practices, the escape produces *institutional mutation*. The society remains itself, but it is no longer identical to the society that existed before contact. The paper argues that this mechanism can be understood as organizational transfer learning: systems can improve not only through retraining model weights but through changes in relationships, protocols, verification practices, and institutional memory.

For financial practitioners, the significance is substantial. Modern financial decisions routinely cross domains: valuation depends on technology, credit depends on legal enforceability, market risk depends on operational resilience, and investment theses increasingly depend on scientific, cyber, or regulatory expertise. A future agentic financial institution may therefore need not only intelligent specialists but also the ability to recognize its own competence frontier, seek external capabilities selectively, negotiate interoperability, preserve provenance, and learn institutionally from the encounter. The paper concludes that the next frontier of autonomous AI is not simply more powerful reasoning. It is the design of *bounded emergence*: environments open enough to permit unexpected organization, yet constrained enough to remain observable, reversible, attributable, and governable.

The problem helps produce the organization that solves the problem. When the organization reaches the frontier of its own competence, the problem may begin to reorganize the boundary itself.

The Capstone Question: What Becomes Autonomous Next?

Students often encounter artificial intelligence as a sequence of increasingly capable models. The progression is familiar: a model that classifies becomes a model that generates; a model that generates becomes a model that reasons; a model that reasons becomes a model that uses tools. At first glance, the natural question seems to be whether the next model will reason better, call more tools, or solve a larger fraction of difficult tasks. Those are important questions, but they are not the central question of this capstone.

The deeper question is architectural: *what object in the system is allowed to change?*

In a conventional model, the surrounding organization is almost entirely fixed. Humans choose the prompt, the data, the available tools, the evaluation criterion, and the handoff to the next stage. Even a highly capable reasoning model remains embedded in a workflow whose structure was supplied from outside. When we introduce an agent, the set of variable objects expands. The agent may choose a next action, decide which tool to call, decide whether to inspect more evidence, or decide when it has enough information to stop. Autonomy has moved from output selection to action selection.

A multi-agent system moves the boundary again. Several agents can divide labor. One can search, another can criticize, another can synthesize. But many multi-agent systems remain orchestrated: the programmer determines the roles, the sequence, and the communication graph. The system contains several intelligent components, yet the organization is still largely designed. This is precisely the distinction emphasized in the broader multi-agent literature: autonomous agents may interact cooperatively or strategically, but the architecture of interaction and the

protocols governing that interaction are themselves first-class design objects [1, 2].

The sequence leading to **ESCAPES** progressively makes those organizational objects variable. Constellations allow multiple agents to become a meaningful working unit. Adaptive systems alter coordination in response to evidence. Endogenous coalitions permit agents to discover collaborators instead of receiving a complete team from the programmer. Constitutional swarms go further by making the rules of collaboration themselves partially endogenous. By the time we arrive at an artificial society, the system has accumulated something that begins to resemble an institutional order: local memory, locally meaningful relationships, conventions for verification, restrictions on information, rules for joining coalitions, and a history of prior interaction.

At that point it is tempting to declare the architecture complete. We have moved from a model to a society. What else is left to make autonomous?

The answer is the **boundary**.

The society still exists inside an experimental container. The programmer decides which agents belong to it, which bulletin board they can see, which institutional rules they inherit, and where their social world ends. The agents may form relationships inside the boundary, but the boundary itself remains an unexamined constant. This is the hidden assumption that NB05 removes.

The paper therefore studies a final conceptual movement: from endogenous organization *inside* a society to selective reorganization *across* societies. The move is subtle. We are not claiming that the system creates its own universe or abolishes all externally supplied constraints. Three societies are still provided. Bulletin boards are still provided. API access, computational limits, logging, safety boundaries, and stopping rules are still supplied by the experimenter. The architecture remains bounded.

What changes is that the system is permitted to recognize when the local organization is insufficient and to treat external collaboration as an admissible organizational response. The finance society does not automatically receive an engineering agent. The programmer does not merge the societies. Instead, the local society must first recognize an unresolved problem, represent the missing capability, discover a foreign communication surface, issue a bounded appeal, attract a foreign responder, confront incompatible rules, negotiate a bridge, and decide what may return home.

This matters because autonomy is no longer located solely in a reasoning loop. It is distributed across a sequence of increasingly social decisions: who is relevant, which relationship should exist, under what rules interaction can occur, which temporary coalition should form, and whether an imported institutional practice belongs in the home society.

For a financial practitioner, that shift is not science fiction. Financial institutions are already organizations of specialized knowledge. A transaction may involve valuation, accounting, tax, law, cybersecurity, industry science, market structure, sanctions, and operations. Human organizations solve such problems by crossing internal and external boundaries: they hire experts, convene committees, engage counsel, negotiate confidentiality agreements, create data rooms, and import practices from other institutions. The notebook asks what a computational analogue of that process might look like when some of the boundary-crossing decisions are made by autonomous agents rather than fully specified in advance.

PEDAGOGICAL LENS

Do not read the notebook as a story about three chatbots. Read it as a controlled experiment in the progressive endogenization of organizational variables. The important output is not the technical answer to the acquisition problem. The important output is the sequence of organizational changes that the problem provokes.

Autonomy Migrates Upward Through the Architecture

The intellectual ladder behind the notebook can be summarized in one sentence: **autonomy migrates upward**. What begins as freedom to choose an answer gradually becomes freedom to choose actions, relationships, coalitions, rules, and eventually selective boundary crossings.

Consider the difference between a model and an agent. A model receives an input and produces an output. An agent exists in time. It can observe, act, receive consequences, maintain state, and choose what to do next. That is already a major architectural transition because the object of optimization is no longer just a response. It is a trajectory. Classic agent theory emphasizes precisely this distinction between passive computation and entities situated in an environment with autonomy over action [2]. Brooks’s argument for intelligence grounded in situated activity also remains relevant here: intelligent behavior can emerge from the organization of interacting activity rather than from a single monolithic representational pipeline [5].

Now add several agents. If the programmer specifies every edge among them, the architecture is richer but still externally organized. The next step is therefore to let relationships become conditional. Perhaps one agent discovers

evidence that another agent is well placed to examine. Perhaps a difficult task causes a temporary team to emerge. The network becomes a function of the problem rather than a fixed diagram.

The notebook calls such temporary, problem-shaped groups **constellations**. The metaphor is useful because the constituent agents remain distinct while acquiring meaning through their pattern of connection. A valuation agent and a strategy agent can exist separately, but when a discovery raises a question about the economic consequences of technical defensibility, the relevant constellation may suddenly include synthesis, valuation, and strategy. The group is not merely a list of components; it is an organizational response to uncertainty.

From there, autonomy moves again. A society can have several constellations, but those constellations operate under rules. Who may see what? What counts as verification? Who receives attribution? Can any agent join? What evidence is sufficient for a decision? When those rules are entirely supplied by the programmer, institutions remain exogenous. Constitutional swarm experiments relax that assumption by allowing agents to participate in rule formation. The resulting artificial society is therefore more than a collection of tools. It possesses something like a local institutional personality.

NB05 begins after that point. Society A, Society B, and Society C are not interchangeable pools of experts. They are different epistemic cultures. Society A behaves like a financial organization. It is sensitive to confidentiality, provenance, economic relevance, and controlled disclosure. Society B behaves like an engineering and scientific organization. It is oriented toward replication, competing hypotheses, peer criticism, and evidentiary openness. Society C behaves like a security organization. It prioritizes need-to-know restrictions, adversarial verification, rapid containment, and chain-of-custody.

These differences are deliberately strong because the capstone depends on them. If all societies shared the same rules, cross-society contact would amount to calling a remote expert. The interesting event occurs only when a useful external capability is embedded in a different institutional order. Then semantic complementarity and institutional incompatibility can coexist.

The final upward migration occurs when the boundary that separates societies becomes part of the decision space. Importantly, the boundary does not disappear. A completely boundaryless system would not illustrate institutional interoperability; it would simply illustrate a larger society. The boundary must remain meaningful for crossing it to matter.

This is why the notebook distinguishes three levels of escape. **Informational escape** occurs when information crosses the boundary. **Relational escape** occurs when a persistent relationship forms across the boundary. **Institutional escape** occurs when a rule, method of organization, or governance idea crosses the boundary and becomes relevant to another society. The progression moves from content to connection to institution.

The concept is closely related to earlier work on distributed problem solving and negotiated task allocation. Smith's Contract Net Protocol showed that distributed nodes can allocate tasks through negotiation rather than centralized command [3]. The capstone pushes the intuition one level upward: not only can tasks be negotiated, but the relationship and even the temporary rule system under which a task is jointly pursued can become the object of negotiation.

We can therefore write the course progression conceptually as

Model → Agent → Constellation → Adaptive Organization → Endogenous Coalition → Endogenous Constitution → Artificial

Each arrow does not mean that the previous object disappears. Models remain inside agents. Agents remain inside constellations. Constellations remain inside societies. Local societies remain inside the ecology. The architecture becomes layered rather than replaced.

That nesting is one of the most important lessons for financial institutions. The move toward more autonomous AI does not necessarily require replacing every existing governance mechanism with one giant general agent. A more plausible trajectory is compositional: bounded intelligent components become embedded in progressively richer structures of coordination, review, delegation, and institutional interaction. The research question is therefore not simply how to build a smarter component. It is how to build systems in which the organization surrounding intelligence can learn without becoming opaque or ungovernable.

Code Unit 1: Building Three Artificial Societies

The first code unit creates the world. This sounds trivial, but the design decision embedded in the world is the foundation of everything that follows. The programmer creates three societies in the same computational environment while preventing the agents from seeing that global environment as a unified social space.

Society A represents finance. Its six specialties—finance, valuation, accounting, ownership, strategy, and synthesis—define the local capabilities available to the acquisition problem. Society B represents technology and engineering, with engineering, patents, materials, manufacturing, scientific validation, and synthesis. Society C represents risk, cybersecurity, and complex systems, with cybersecurity, anomaly detection, network risk, adversarial analysis, resilience, and synthesis.

The societies also receive different constitutions. These constitutions are not decorative metadata. They encode operational principles that later shape coalition formation and cross-border friction. Society A follows a *Confidential Capital Protocol*. Society B follows a *Replicable Inquiry Charter*. Society C follows an *Adversarial Containment Covenant*. Each constitution says something different about sharing, verification, attribution, restrictions, and coalition formation.

The critical initial condition is informational asymmetry between the experimenter and the agents. The Python program knows that all three societies exist. The agents do not. Each agent's list of known bulletin boards initially contains only its home board. There is no global bulletin board, no exposed global agent directory, and no central router. That separation prevents the experiment from quietly solving the boundary problem before it begins.

The cell also creates the ledgers that will later make emergence inspectable. There is an event ledger, an escape ledger, an institutional-transfer ledger, a compact ledger, a cross-coalition ledger, and a collision ledger. These structures may look like bookkeeping, but pedagogically they embody a strong governance principle: **unexpected organization is allowed; invisible organization is not**. If a relationship crosses societies, it should leave a trace. If a compact is formed, its terms should be inspectable. If an institutional rule migrates, its source and adaptation should be attributable.

Claude is also configured in this cell, but in a deliberately subordinate role. A reusable helper makes structured calls, strips accidental code fences, validates required fields, and falls back deterministically when the model is unavailable or returns malformed output. The maximum number of calls is bounded. This architecture is important because the notebook is not trying to demonstrate that “Claude controls the society.” It is trying to demonstrate a hybrid design in which deterministic code handles known rules while a language model is used where interpretation, analogy, negotiation, or semantic judgment genuinely matter.

For students coming from traditional quantitative finance, this separation is worth emphasizing. We often think of determinism and generative AI as competitors: either a system is rule-based or it is model-driven. In well-designed autonomous architectures, the two can be complementary. Counting, thresholding, graph construction, provenance, and acceptance logic can remain explicit. Semantic interpretation can be delegated to an LLM. The boundary between those responsibilities is itself part of system design.

The first cell therefore does not yet produce emergence. It produces *conditions under which emergence can be observed*. Three societies exist, but they are not an ecology. They are three disconnected institutional worlds.

Code Unit 2: Local Life Before Any Escape

The second code unit is designed to prevent a misleading conclusion. If the local societies were incapable of accomplishing anything, then external search would simply be a repair mechanism. The notebook instead insists on a period of successful local life. Each society must first demonstrate that bounded organization can work.

Agents search according to specialty. The findings are synthetic, but they are structured to resemble the kinds of partial discoveries that appear in real professional work. A finance agent may conclude that revenue quality exceeds peers but leave unresolved the durability of pricing power. A valuation agent may find a positive acquisition spread while asking how technical risk should enter the discount rate. A patent agent may find broad claims that overlap with prior art. A manufacturing agent may identify an improved process window while questioning stability across suppliers. A cybersecurity agent may discover a privileged account in an unexpected execution path without yet knowing whether the behavior is benign automation or lateral movement.

The pattern matters. Each discovery contains both a finding and an unresolved question. Intelligence progresses by reducing uncertainty, but valuable inquiry often reveals new uncertainty at the same time. This makes the environment suitable for later coalition formation because discoveries can become invitations for complementary expertise.

Each evidence object carries provenance, society, agent, specialty, confidence, novelty, semantic value, and the unresolved question. It is added to the society's discoveries, the agent's local memory, and the local bulletin board. The board therefore becomes an institutional memory surface rather than a one-off messaging channel.

Local responses remain deterministic. An agent responds when it has sufficient competence overlap and the discovery has sufficiently high semantic value. The design deliberately avoids using Claude for every local interaction. That choice keeps the mechanism inspectable and avoids conflating autonomy with stochastic language generation. It also

reinforces a more general lesson: AI systems become more trustworthy when open-ended model judgment is used selectively rather than indiscriminately.

Students should observe that the three societies already display useful collective intelligence. Society A can build a coherent acquisition view. Society B can generate increasingly disciplined technical hypotheses. Society C can assemble a map of vulnerabilities and resilience questions. Nothing in the experiment says that every difficult problem should be escalated outside the boundary. Indeed, the future ecology would be dysfunctional if every local uncertainty triggered external collaboration.

This gives us a necessary principle for autonomous organizations: **local sufficiency should be preferred when local competence is adequate**. Escape becomes meaningful only when an unresolved issue is both important and beyond the local competence frontier.

In organizational terms, this resembles the difference between a firm that outsources everything and a firm that knows when specialist external expertise is genuinely required. The ability to remain local can be as important as the ability to cross boundaries. A system that always collaborates externally has no meaningful institution of its own; it is simply a thin routing layer.

The code unit therefore establishes the baseline against which escape can later be interpreted. We see local search, local communication, local response, local memory, and local relationship formation. The boundary remains intact because the society is still capable of productive life within it.

Code Unit 3: Local Constellations and Institutional Personality

The third code unit transforms local interaction into local organization. The distinction is important. A society can contain many messages without containing meaningful structure. Organization appears when relationships begin to form patterned working arrangements around problems.

The notebook calls these arrangements local coalitions or constellations. It sorts discoveries by semantic value, selects the most promising ones, identifies the originating agent, and then chooses collaborators according to the society's constitutional logic. Society A forms small, economically relevant, controlled-sharing coalitions. Society B favors broader specialty diversity in order to support replication and criticism. Society C favors compact teams consistent with need-to-know principles.

The same number of agents can therefore produce different network topologies because the institutions differ. This is the first point in the capstone where students can see a direct causal link between rules and organization. The societies are not different merely because they contain different data. They are different because their constitutions define what kinds of relationships are legitimate and useful.

The idea has a close analogue in institutional economics. Ostrom's work on self-governing institutions showed that durable collective action cannot be understood only by examining individual preferences or resources; rules-in-use shape who participates, what is monitored, how obligations are interpreted, and how collective outcomes are sustained [4]. The notebook translates that intuition into a deliberately simplified computational setting. We do not reproduce the complexity of real political institutions, but we make one powerful point visible: collaboration is rule-dependent.

This is where the metaphor of a *society* becomes more than rhetoric. Society A and Society B may contain equally capable LLM-backed agents, yet those agents do not live in identical social worlds. One world expects confidentiality. Another expects reproducibility. One favors economic materiality. Another favors falsifiable claims. A third expects compartmentalization and adversarial challenge. As a result, the same external request can be interpreted differently depending on the institutional environment in which it arrives.

The notebook visualizes the local graphs independently. This is pedagogically useful because students can see that each society possesses a coherent interior before any cross-border edge exists. The graph is not yet an ecosystem graph; it is three graphs. That distinction will later make the first cross-society relationship visually and conceptually important.

The metrics remain intentionally simple: number of relationships, coalition count, specialty diversity, unresolved problems, and the society's signature rule. We do not need sophisticated graph theory to make the lesson. The point is not to maximize network density. It is to understand why the network has the shape it does.

A subtle idea appears here: **the constitution does not merely constrain agents; it also enables them**. Finance confidentiality can prevent leakage while permitting economically useful collaboration. Engineering replication rules can increase confidence in claims. Cybersecurity containment can allow sensitive threat analysis to proceed without unrestricted disclosure. Governance is therefore not simply friction. It is part of the productive architecture.

This becomes essential later. When two societies collide institutionally, the objective will not be to remove all governance so that agents can collaborate freely. The objective will be to create a bridge that preserves the enabling functions of both local institutions.

Code Unit 4: The Local Competence Frontier

The competence frontier is the conceptual hinge of the entire notebook. Without it, escape would be arbitrary. The programmer could simply decide that at time step five an agent is allowed to contact another society. That would create cross-society traffic, but not meaningful organizational autonomy.

Instead, the system must first recognize a particular kind of condition: *the question matters, we understand why it matters, and we do not possess the capability required to resolve it*. That is a richer state than ignorance.

Society A provides the initiating example. Its finance, valuation, accounting, ownership, strategy, and synthesis agents have built an attractive acquisition thesis. The economics look promising. Yet the value of the acquisition depends on whether the target’s technology is defensible, independently reproducible, and scalable. The local agents can understand the economic consequences of those properties. They cannot reliably determine whether those properties are technically true.

The code compares required external-looking capabilities—engineering, patents, scientific validation, and manufacturing—against the maximum competence available inside Society A. That deterministic comparison provides an initial structural signal. But the notebook then uses Claude for a deeper semantic judgment. The model is asked to return importance, confidence, the unresolved issue, the missing capability, the expected value of external help, and a reason the local society is insufficient.

This is precisely the kind of task for which language-model judgment is appropriate. It is not arithmetic. It is a meta-level interpretation of the relationship between a problem and an organization’s capability set. The system is asking a question that human professionals ask constantly: “Do we know enough to decide, or are we outside our competence?”

The notion has important implications for financial institutions. Many catastrophic errors do not arise because organizations lack intelligence. They arise because organizations fail to represent the boundary of their intelligence. A valuation team may understand a model perfectly while misunderstanding the science on which the model depends. A credit committee may understand cash flows while underestimating legal enforceability. A risk team may understand price distributions while missing operational fragility. Competence is domain-specific, and autonomous systems should be able to represent that fact rather than treat model confidence as universal expertise.

The code makes a deliberate pedagogical choice: the escape still does not happen. First the frontier is logged and displayed. The output explicitly says that the frontier is visible but no escape has occurred. This separation matters because recognition and action are different capabilities. A system might correctly recognize that it lacks competence and still decide that external search is too costly, too risky, or unnecessary. In later research, such decisions could incorporate explicit economics. In this notebook, the key point is simpler: the society can represent its limit before it crosses it.

That ability is a higher-order form of autonomy. The society is no longer merely selecting among actions inside a given organization. It is reasoning about the adequacy of the organization itself.

REPRESENTED IGNORANCE

A competence frontier is not “the model is uncertain.” It is the organizational statement: *this issue is decision-relevant, our local capabilities are insufficient, and we can characterize the type of capability that is missing*. Represented ignorance is therefore a governance asset.

Code Unit 5: The First Escape

The first escape is the event that changes the ontology of the experiment. Before it occurs, there are three artificial societies. After it occurs, there is at least the possibility of an ecology.

The term *escape* must be handled carefully. In everyday language it suggests loss of control, evasion, or a system breaking free. That is not what the notebook means. Escape is defined technically as a boundary-crossing event: an agent operating inside one endogenous local organization initiates interaction inside a previously external agentic environment.

Society A’s synthesis-oriented agent is chosen because it is well positioned to represent the integrated acquisition problem. The agent then learns one fact that it did not previously know: `BOARD_B` exists. It still does not know who belongs to Society B. It receives no directory of engineering agents. It is not routed to a patent specialist. The system exposes only a communication surface.

This design echoes a powerful coordination idea in both human institutions and distributed systems. Addressability can be more important than omniscience. One does not always need to know exactly who can help. It may be enough

to know where a relevant appeal can be made visible. Smith's Contract Net Protocol formalized task announcement and bidding as a means of distributed allocation [3]; NB05 uses a looser semantic analogue in which an appeal is posted to an unfamiliar institutional environment and potential responders judge relevance themselves.

Claude is used to help formulate the appeal. The request explains what is known, what is unknown, why the missing knowledge matters, and what kind of expertise may be useful. At the same time, it preserves Society A's constitutional commitments by withholding unnecessary transaction-sensitive identifiers. This is important. The escape is not permission to abandon the home constitution. The foreign message is already an exercise in bounded interoperability.

The notebook records the event as **ESCAPE_001**. The record contains the escaping agent, home society, foreign society, originating problem, competence gap, foreign board, timestamp, message, and institutional status. The ledger therefore treats boundary crossing as a first-class governance event.

Why is the first escape so important? Because before it, the separation among societies is absolute from the agents' perspective. After it, information created inside one institutional world exists inside another. The computational graph has not yet gained a persistent cross-society relationship, but the informational topology has changed.

This is **informational escape**. It is the shallowest of the three escape levels, but it is still a major transition. A local problem has generated an action whose target lies outside the local organization.

For students, the most useful way to interpret the event is to imagine a financial analyst who recognizes that a technology question cannot be answered internally and posts a carefully de-identified request to an external scientific network. The analyst does not yet have a consultant. There is not yet a contract. There is only an externalized question. That question is now capable of attracting intelligence that the originating organization did not possess.

The notebook's deeper proposition can now be stated more sharply: **the problem is beginning to reshape the organization's boundary**. The acquisition problem first shaped local collaboration. Now it has become strong enough to push intelligence beyond the institution in which the problem originated.

Code Unit 6: Foreign Contact and Semantic Matching

An informational escape does not automatically create collaboration. The foreign society may ignore the message. No agent may find it relevant. A responder may lack sufficient confidence. If the notebook simply routed the request to the correct specialist, the central problem would disappear: the programmer would have performed the organizational intelligence on behalf of the agents.

The sixth code unit therefore asks each plausible Society B specialist a semantic question: *does anything I know make me useful to this unfamiliar problem?*

This is where the revised pedagogical version of the cell is especially important. In the first version, Claude's relevance judgment could be conservative enough that no foreign agent qualified. That outcome is legitimate in a research simulation, but pedagogically it can obscure the concept the notebook is designed to teach. The revised cell therefore combines a deterministic specialty prior with Claude's semantic interpretation. Scientific validation, engineering, and patents receive high structural priors because the foreign request explicitly concerns reproducibility, technical plausibility, and defensibility. Claude still judges the actual message and proposes a contribution. The final relevance score blends those two sources rather than allowing one stochastic model call to erase an otherwise obvious capability match.

This is not the same as central routing. No agent is preselected as the winner. Several agents can independently qualify. Scientific validation might offer a replication protocol. Engineering might evaluate physical and architectural plausibility. Patents might examine prior art. Manufacturing might assess scale-up risk. Materials might recognize hidden degradation constraints. The system then ranks the qualifying responses and creates a lead cross-society relationship.

A pedagogical safeguard remains in place if every semantic judgment rejects the request despite an overwhelming structural specialty match. This safeguard is explicitly labeled. In a production research setting, one might prefer to let the failed match stand and analyze it. In a teaching notebook, the purpose is different: students need to observe what relational escape looks like so that later institutional collision can be studied. The safeguard therefore makes the didactic mechanism visible rather than pretending that the demonstration emerged with no scaffolding.

Once a responder qualifies, the notebook creates a persistent relationship between the escaping Society A agent and the lead Society B responder. This is **relational escape**. The world is now connected not merely by a message but by a social edge.

The distinction is conceptually important. Information can cross without creating durable coordination. A research paper can be read by another organization without the organizations forming a relationship. Relational escape requires something more: the receiving agent recognizes relevance, proposes a contribution, states conditions for

participation, and becomes part of a continuing interaction.

The resulting relation has strength, source society, target society, reason, scope, and a trigger escape. These are deliberately simple constructs, but they make the emerging network inspectable.

For financial practitioners, semantic matching may prove one of the most consequential capabilities of future agentic systems. Most real problems do not arrive pre-labeled with the name of the correct expert. A distressed credit may contain a supply-chain issue. A fraud pattern may resemble a cybersecurity anomaly. A transaction may hinge on scientific evidence that no finance ontology anticipated. Language models can potentially recognize those cross-domain analogies more flexibly than fixed routing taxonomies.

The tradeoff is governance. Semantic flexibility increases the chance of useful unexpected connections, but it also makes behavior less predictable *ex ante*. The appropriate response is not to eliminate semantic judgment. It is to surround it with thresholds, provenance, bounded candidate sets, call budgets, structured outputs, and explicit fallbacks.

That hybrid architecture is one of the paper's broader design themes: **deterministic mechanisms define the safe field of play; language models exercise judgment inside it.**

Code Unit 7: Institutional Collision

At this point the two societies have discovered something uncomfortable: technical complementarity does not imply institutional compatibility.

Society A wants help. Society B can provide help. Yet the rules that made each society locally coherent are different. The finance constitution favors minimal necessary disclosure, confidentiality, provenance, and economic relevance. The engineering constitution favors enough disclosure to permit independent replication, peer criticism, and competing hypotheses. One institution is designed to protect a transaction. The other is designed to challenge a technical claim.

Neither orientation is obviously superior. Indeed, each would be defective if it simply adopted the other's logic wholesale. A finance society that disclosed all commercial context in the name of scientific openness could destroy transaction confidentiality. An engineering society that accepted opaque claims because they were commercially sensitive could no longer provide meaningful independent verification.

This produces COLLISION_001, the first institutional collision.

Claude receives the two constitutions and is asked to identify compatible rules, conflicting rules, unresolved governance questions, collaboration risk, and possible interoperability principles. The prompt explicitly forbids selecting one constitution as superior. That instruction matters. The problem is not constitutional competition. It is institutional interoperability.

Some rules are naturally compatible. Both societies value provenance. Both require verification before a material claim should drive a decision. Both care about attribution. Those commonalities provide a foundation for cooperation.

The conflicts are equally important. How much detail must be shared? Who may join the review? Can a Society B scientist invite another reviewer when Society A's confidentiality rules favor narrow participation? Can derived artifacts return to the home societies? Who owns jointly produced conclusions? What happens if replication requires information that Society A is not authorized to disclose?

Notice how far we have moved from a conventional AI task. The original question was whether a technology was defensible. Now the system is reasoning about jurisdiction, disclosure, attribution, membership, and return rights. The technical problem has generated an institutional problem.

This is precisely why the capstone belongs in a course for financial practitioners. Financial work is full of such collisions. Investment banks interact with lawyers, auditors, engineers, consultants, regulators, and counterparties whose professional obligations differ. The solution is rarely to erase those obligations. Instead, real institutions create bounded interfaces: engagement letters, confidentiality agreements, data rooms, clean teams, Chinese walls, reliance language, expert mandates, and committee charters.

NB05 constructs a primitive computational analogue. The institutional collision makes explicit that cross-society collaboration requires more than semantic matching. A capability can be useful and still be inaccessible under current rules.

This insight also clarifies the nature of advanced autonomous systems. The frontier is not simply building an AI that "knows what to do." It is building systems that can represent the rule conflicts surrounding what they want to do. The more organizational autonomy we permit, the more important institutional reasoning becomes.

Ostrom's analysis of self-governance is useful here because it reminds us that robust cooperation depends on rules that fit local circumstances rather than on one universal institutional form [4]. The notebook therefore resists the temptation to merge constitutions. Diversity is not a bug to be optimized away. It is part of the ecology's resilience

and specialization.

Code Unit 8: Negotiating a Cross-Society Compact

The eighth code unit asks the question that follows naturally from institutional collision: if neither local constitution should dominate, what new organizational object can make collaboration possible?

The answer is a **cross-society compact**.

The compact is deliberately narrow. It is not a new universal constitution. It does not merge the societies. It does not create a sovereign supervisor above them. It is a temporary bridge agreement tailored to the current task. This makes it analogous to a treaty, an engagement protocol, a joint operating agreement, or a special-purpose jurisdiction.

Claude proposes the compact because the task requires semantic synthesis across partially conflicting rule systems. The resulting object specifies permitted information exchange, confidentiality, verification, attribution, decision rights, duration, joining conditions, exit conditions, provenance, and return rights. Each clause exists because a conflict identified in the previous cell requires a concrete resolution.

The permitted exchange is de-identified. Technical claims, test artifacts, architecture constraints, and claim charts can move, while deal identities and unnecessary commercial terms remain protected. Society B is allowed to perform independent replication and adversarial review on disclosed evidence. Society A retains acquisition decision authority. Provenance follows every exchanged artifact. Additional members require a documented capability need and approval from representatives of both societies. Either side may exit. Restricted raw artifacts do not automatically return home.

The notebook then refuses to let Claude's compact become law simply because it was eloquently written. Acceptance is deterministic. Society A checks for confidentiality, provenance, and controlled sharing. Society B checks for verification, challenge, and technical access. Only if both societies' non-negotiable principles are visibly satisfied can the compact support a coalition.

This is an important governance pattern for LLM-enabled institutions: **generative proposal, deterministic admissibility**. A language model can invent a nuanced institutional bridge, but explicit rule checks determine whether that bridge enters the operative system.

If both sides accept, **CROSS_COALITION_001** is formed. This coalition is conceptually different from the local constellations of Code Unit 3. It spans societies. It is therefore a **meta-constellation**: a temporary organizational form whose members remain citizens of different local institutions but collaborate under a third, task-specific governance layer.

The meta-constellation does not erase local identity. The Society A agent remains accountable to Society A. The Society B agent remains accountable to Society B. The compact defines the narrow shared jurisdiction between them.

This architecture may prove highly relevant to real financial organizations. Consider a bank that uses an external AI-enabled scientific diligence service. A naive architecture might simply transmit a document and receive an answer. A more institutionally serious architecture would specify data minimization, permissible inference, auditability, verification rights, retention, model provenance, joining conditions, and authority boundaries. In other words, it would create a compact.

The significance of the cell is therefore larger than its synthetic case. It demonstrates that autonomous systems can potentially create *temporary institutions* in response to problems while remaining inside constitutional constraints.

That capability sits near the frontier of agentic design. It requires a system to reason not just about tasks and tools, but about the social contract under which tasks and tools can be shared.

Code Unit 9: Return, Transfer, and Institutional Mutation

If the story ended when the engineering society answered the finance society's question, the experiment would still be interesting, but it would not yet demonstrate the deepest form of self-improvement.

The coalition produces a technical conclusion: the technology appears plausible and partly reproducible, but defensibility depends on a narrow combination of implementation know-how and patent claims, while manufacturing scale remains a material diligence risk. Society A now has better information. That is the obvious value of the interaction.

But the returning agent can bring home three different classes of transfer.

The first is **informational transfer**. New factual knowledge crosses the boundary: what is likely true about the technology and what risks remain.

The second is **capability transfer**. Society A learns a method: independent replication plus explicit competing-hypothesis review for decision-critical technical claims. The method is reusable beyond the current target.

The third is **institutional transfer**. Society A encounters a rule embedded in Society B's scientific culture: separate claim origination from independent verification when technical evidence can materially change a decision.

The third transfer is the most profound because it can change the future organization of work. Claude evaluates whether the imported institutional idea is relevant to Society A without replacing Society A's constitution. The recommended adaptation preserves transaction confidentiality while requiring an independent verification memo for decision-critical technical claims. Society A's agents then vote according to deterministic support rules. A two-thirds threshold determines adoption.

If the proposal passes, the notebook records `INSTITUTIONAL_MUTATION_001`. The ledger stores the source society, destination society, original rule, imported idea, local adaptation, acceptance mechanism, and status.

The word *mutation* is intentionally provocative but should not be misunderstood. Society A does not become Society B. Its identity persists. The imported rule is adapted to the receiving institution's context. This is exactly what makes the event organizational learning rather than institutional replacement.

The deeper idea is that AI systems can improve without updating model weights. Much contemporary discussion treats learning as synonymous with training. Yet organizations learn in many other ways. A firm learns when it changes a review process. A committee learns when it adds an independent challenge function. A trading desk learns when it modifies an escalation rule. A bank learns when it creates a new data lineage requirement after an incident.

The notebook therefore introduces the idea of **organizational transfer learning**. What transfers is not a parameter vector but an institutional pattern.

This has major implications for how we think about self-improving AI. A self-improving system need not be a model that rewrites its own code. It can be an organization of models and agents that changes its relationships, protocols, verification rules, and memory structures as a consequence of experience. The object of improvement is the architecture.

That framing is especially appropriate in finance, where uncontrolled self-modification of core models would often be undesirable, but controlled improvement of workflows, review structures, and escalation protocols is central to institutional learning.

The notebook's architecture remains bounded. Mutation is logged. Provenance is preserved. Adoption is explicit. The system is permitted to change, but the change is not invisible.

The deepest transfer is not an answer. It is a better way of deciding which answers deserve trust.

Code Unit 10: From Three Societies to an Ecology

The final code unit expands the world cautiously. The objective is not to maximize interaction. Dense connectivity would actually weaken the conceptual claim because an ecology of autonomous systems is not a universal chat room. It is a set of locally coherent societies capable of selective interdependence.

The notebook therefore introduces a small number of additional competence-frontier opportunities. Society B may need adversarial-analysis expertise from Society C when evaluating whether a technically strong architecture is exploitable. Society C may need strategic-finance interpretation from Society A when a resilience intervention must be prioritized under capital constraints. Society A may later seek cybersecurity expertise from Society C when acquisition value depends on whether an operational vulnerability is containable.

Each potential escape is gated. The originating problem must have sufficiently high value. Local competence in the required specialty must be inadequate. The destination society must possess sufficiently strong competence. Only then does the home agent discover the foreign board and initiate an informational escape.

Even after that, relational escape is conditional. A foreign specialist must have sufficiently high competence to respond. When constitutions differ, the notebook creates a minimal inspectable compact before a cross-society coalition forms. The mechanics are simplified, but the principle is consistent with the first A-to-B encounter: boundary crossing remains selective and governed.

The final graph is the visual culmination of the notebook. Each society remains identifiable as a cluster. Local relationships remain visible. Cross-society edges appear only where relational escape has actually occurred. The societies have not dissolved into a single population.

The dashboard then counts local collaborations, escapes, informational escapes, relational escapes, cross-society coalitions, institutional collisions, temporary compacts, institutional transfers, and institutional mutations. These are

not merely simulation statistics. They are the observables of a new level of organization.

At the beginning of the notebook, the relevant object is an agent. By the end, the relevant object is an ecology. The ecology contains agents, local constellations, societies, cross-society relationships, and temporary compacts simultaneously. Intelligence has become multi-level.

This is why the notebook is a capstone rather than another example of multi-agent orchestration. It demonstrates a progression from **local autonomy** to **inter-organizational autonomy**. The system can now represent the possibility that the best organization for solving a problem may not lie entirely inside the current institutional boundary.

Yet the architecture remains explicitly anti-centralizing. No global coordinator has appeared. No universal constitution has been imposed. No omniscient agent knows the entire ecosystem. The ecology is produced through selective bridges among locally autonomous societies.

That distinction is essential. A centralized super-agent connected to every specialist might solve the same technical tasks. But it would teach us much less about how autonomous organizations could coexist, negotiate, and learn from one another.

The final state is therefore not “one larger AI.” It is a primitive institutional ecosystem.

Three Scales of Intelligence: Individual, Collective, and Meta-Constellation Dynamics

The notebook becomes easier to understand when we separate three levels of dynamics that coexist throughout the experiment.

At the **individual level**, each agent has a specialty, a competence profile, local memory, discoveries, relationships, known bulletin boards, and institutional memory. The individual agent is the locus of specialized judgment. A valuation agent interprets economic value. A scientific-validation agent interprets replicability. A cybersecurity agent interprets threat patterns. Individuals encounter evidence through the lens of their capabilities.

Individual agency matters because the larger structures are not meaningful if the components are interchangeable. The scientific-validation agent is useful precisely because Society A does not already possess that competence. Heterogeneity creates the possibility of complementarity.

At the **collective level**, local societies transform separate specialist judgments into organized intelligence. Bulletin boards make discoveries public inside the local boundary. Relationship formation creates a memory of useful interaction. Coalitions create temporary divisions of labor. Constitutions shape how those coalitions form and what they are allowed to do. The society is therefore not just the sum of its agents. It contains organizational state that no individual agent fully represents.

This distinction parallels long-standing multi-agent research, which treats interaction structure as a source of system-level behavior rather than merely a transport layer between independent programs [1]. It also resonates with distributed-control traditions in which intelligence emerges from coordinated activity rather than a single central representation [5].

At the **meta-constellation level**, the object of organization spans societies. A cross-society coalition contains agents whose home institutions remain distinct. The coalition therefore has a double character: it is a working group and an interoperability regime. The compact defines the temporary jurisdiction within which the group can act.

This level is genuinely new. Local constellations answer the question, “Which agents inside our society should work together?” Meta-constellations answer the question, “Which societies need to connect, through which representatives, and under what temporary rules?”

The three levels can be thought of as nested forms of problem-induced organization:

Individual judgment → Local constellation → Cross-society meta-constellation.

Each level has its own memory. Individuals have local memory. Societies have bulletin boards, relationship graphs, and local rule histories. Meta-constellations have compacts, provenance chains, and cross-society ledgers. Each level therefore accumulates experience differently.

This nesting also changes the meaning of failure. An individual may fail because it lacks knowledge. A local constellation may fail because the society lacks a specialty. A meta-constellation may fail because institutions cannot agree on an admissible bridge. The architecture helps us distinguish epistemic insufficiency from organizational insufficiency and institutional incompatibility.

For finance, that distinction is valuable. A failed investment process may not be caused by a bad model. It may be caused by a missing expert, an inappropriate committee, a confidentiality barrier, a poorly designed escalation path, or an institutional rule that prevents relevant evidence from reaching the decision maker. Agentic systems should eventually be capable of diagnosing these different failure modes rather than treating every problem as another prediction task.

The meta-constellation concept also suggests a different approach to enterprise AI. Instead of building one monolithic agent that owns every tool and data source, firms could maintain locally governed agentic domains—finance, legal, cyber, operations, research—and allow them to create temporary governed bridges when problems require cross-domain intelligence. This architecture could preserve specialization and authority boundaries while still permitting emergent collaboration.

The challenge, of course, is that meta-level emergence is harder to predict. That is why the notebook pairs increasing autonomy with increasing observability. As organization becomes more endogenous, ledgers, provenance, compacts, and explicit acceptance criteria become more important rather than less.

Agent Motivation: Mission-Driven, Not Economically Self-Interested

One question naturally arises once we begin describing societies and coalitions: what motivates the agents to participate? Do they receive compensation? Do they maximize utility? Do they bargain over rewards?

In this notebook, the answer is deliberately no.

The agents are simple goal-directed specialists rather than fully economic actors. Each agent has a domain-specific objective and behaves as if success consists of making a useful contribution to the problem confronting its society while respecting institutional rules. A valuation agent is oriented toward evaluating the acquisition. A scientific-validation agent is oriented toward assessing whether a claim can survive independent scrutiny. A cybersecurity agent is oriented toward identifying and containing relevant risks.

There is no payment mechanism, wealth account, bounty, Shapley allocation, negotiated fee, or explicit utility function. Relationship strength is not compensation. Coalition membership is not compensation. Institutional voting is not compensation. When we say the agents are “motivated by success,” we mean this operationally: the code gives them roles, relevance criteria, thresholds, and goals that cause them to contribute when their capabilities fit the problem.

This simplification is pedagogically important. Incentives are a profound subject in their own right. If economic self-interest were introduced at the same time as endogenous relationships, endogenous institutions, and cross-society compacts, it would become difficult to know what mechanism produced the observed behavior. By keeping motivation mission-driven, the notebook isolates the organizational phenomenon.

Future work could add compensation as another endogenous layer. External agents might bid for tasks. Societies might share surplus. Reputation might affect future coalition invitations. Institutional ideas might generate continuing credit. At that point the ecology would begin to resemble an economy rather than only a social system. But that is a different experiment.

For the present capstone, the cleaner interpretation is this: agents are members of purpose-driven organizations. Their primary objective is successful task completion under institutional constraints. The research question is how such agents reorganize relationships and boundaries when local competence becomes insufficient.

Self-Improvement Without Retraining: The Organization Learns

The notebook belongs to a broader sequence about self-improving systems, but its notion of self-improvement differs from the most familiar one. We are not primarily studying a model that rewrites its own weights, modifies its source code, or recursively optimizes its reasoning algorithm. We are studying a system whose *organization* changes as a result of experience.

This distinction is important because organizations can become more capable without any individual member becoming more intelligent. A committee can improve by changing its review order. A hospital can improve by introducing an escalation protocol. A trading desk can improve by separating signal generation from risk approval. A bank can improve by adding independent model validation. None of those changes require a person to acquire a new brain. The architecture surrounding decision making has learned.

NB05 makes this idea computationally visible. Before the escape, Society A does not have a specific institutional rule requiring independent technical verification of decision-critical claims by an agent separate from the originator. After interacting with Society B, it may adopt an adapted version of that rule. Future problems will therefore be processed differently even if every underlying agent remains unchanged.

That is a legitimate form of self-improvement.

It also suggests a taxonomy of learning in agentic systems. **Parametric learning** changes model weights. **Memory learning** changes what the system can retrieve from prior experience. **Relational learning** changes who collaborates with whom. **Procedural learning** changes the sequence or workflow through which tasks are processed. **Institutional learning** changes the rules governing collaboration and authority. **Ecological learning** changes how independently organized systems connect and transfer practices.

Traditional machine learning focuses primarily on the first category. Modern agent systems increasingly use the second and third. The capstone pushes deliberately toward the fourth, fifth, and sixth.

This perspective is consistent with the broader pedagogical arc represented in Reynoso's repositories on autonomous systems, agentic architectures, governance-first systems, and course materials [6, 7, 9, 10]. Across those materials, the recurring theme is that capability should be understood at the level of the system rather than only at the level of a model.

The idea also clarifies why provenance is central. If the organization is allowed to learn by changing rules, then we need to know which experience produced the change. Otherwise institutional adaptation becomes indistinguishable from drift. NB05 therefore records the source society, destination society, original rule, imported idea, adaptation, and acceptance mechanism.

This is the organizational analogue of experiment tracking. We would not want a production model to change parameters without a record of training data and evaluation. Similarly, we should not want an autonomous institution to change governance rules without a record of why the rule changed and who authorized the change.

For financial institutions, this form of learning may be more plausible in the near term than unrestricted recursive self-modification. Regulated organizations already change policies, controls, and committee structures in response to experience. An AI-enabled system that proposes such changes, evaluates them, and records adoption could be both powerful and governable if authority remains properly bounded.

The self-improving object in NB05 is therefore not merely the agent. It is the **society**.

Governance: From Exhaustive Control to Bounded Emergence

The notebook deliberately increases the system's capacity to surprise its designer. That is the source of its power and its governance challenge.

If every interaction were fully specified in advance, there would be no meaningful emergence. The programmer would know exactly which agent contacts which specialist under which rule. The architecture would be dynamic only in the superficial sense that code branches execute conditionally. By contrast, semantic matching, foreign discovery, temporary compacts, and institutional transfer make some future organizational states genuinely contingent on what the system encounters.

The appropriate governance response is not to eliminate that contingency. Doing so would eliminate the phenomenon we want to study. Instead, governance must move up one level.

The paper uses the term **bounded emergence** for this design philosophy. The system should be open enough for unexpected useful organization to appear, but constrained enough that the organization remains observable, reversible, attributable, and governable.

Observability means that important events leave records. Escapes are logged. Cross-society relationships are logged. Collisions are logged. Compacts are inspectable. Transfers preserve provenance. Mutations record their acceptance mechanism.

Reversibility means that temporary bridges have exit conditions. The compact does not create permanent irreversible authority merely because one interaction was useful. Local societies can terminate participation.

Attribution means that evidence and institutional ideas retain a source. An artifact does not become anonymous simply because it crosses a boundary. Provenance travels with it.

Bounded authority means that no agent acquires global sovereignty. The Society B responder can validate technical claims but does not acquire acquisition-decision rights. Society A can make the acquisition decision but cannot unilaterally redefine Society B's scientific standards. The compact partitions authority rather than pooling it indiscriminately.

Local identity means that interoperability does not imply constitutional convergence. Society A remains finance. Society B remains engineering. Society C remains cybersecurity and risk. Diversity persists after contact.

These principles echo a larger tradition in institutional analysis. Ostrom's work showed that self-governance does not mean absence of rules; it means rule systems adapted to local collective-action problems, with monitoring, boundaries,

and legitimate processes for modification [4]. The notebook's institutions are far simpler, but the lesson is similar: emergence without governance is not the same as self-governance.

For students in finance, this is perhaps the most important translation. Autonomous AI should not be equated with "letting the model do whatever it wants." In serious organizations, increased autonomy must be paired with stronger architectural control. The control changes form. Instead of scripting every action, we specify admissible domains, authority boundaries, audit surfaces, escalation rules, and constitutional constraints.

The revised Code Unit 6 illustrates the same philosophy at a micro level. Semantic matching is open-ended, but candidate specialties are bounded, call budgets are bounded, outputs are structured, thresholds are explicit, and a pedagogical safeguard is visible. Code Unit 8 does the same at an institutional level: Claude proposes a compact, but deterministic acceptance rules decide whether the compact is admissible.

The general architecture can therefore be expressed as

Open-ended judgment inside explicit institutional boundaries.

This is likely to be a central design pattern for autonomous systems in regulated environments. The model can be creative about how to solve the problem, while the system remains strict about what kinds of authority, data movement, commitments, and irreversible actions are permitted.

The capstone therefore argues for a shift in governance philosophy. When systems are simple, control can be procedural: specify the steps. When systems become adaptive, control must become constitutional: specify the boundaries within which steps may be invented.

Why This Matters for Financial Institutions

The synthetic acquisition case is deliberately chosen because finance provides a natural laboratory for cross-domain autonomy. Financial decisions are rarely purely financial. Their value depends on facts and institutions that sit outside traditional financial expertise.

An acquisition model may be numerically flawless yet economically useless if the target's technology cannot scale. A credit model may estimate default probability accurately while ignoring that collateral is legally unenforceable. A trading strategy may produce attractive backtests while depending on operational infrastructure vulnerable to cyberattack. A structured product may satisfy pricing logic while violating suitability or disclosure constraints. A sovereign-risk analysis may miss a supply-chain bottleneck understood by engineers rather than economists.

Human financial institutions already solve these problems through organizational boundary crossing. Deal teams engage counsel, accountants, technical advisers, cybersecurity specialists, tax experts, environmental consultants, and industry specialists. They create data rooms and confidentiality agreements. They distinguish decision rights from advisory rights. They preserve reliance chains. They form temporary committees. They disband those committees when the transaction ends.

NB05 asks what it would mean to make parts of that architecture agentic.

The first requirement is not a more powerful valuation model. It is the ability to recognize a competence frontier. A future AI financial practitioner should be able to say: "The economic sensitivity is high, but the uncertainty is technical and lies outside the current agent constellation." That diagnosis is more valuable than producing another confident financial answer.

The second requirement is selective external search. The system should not expose a universal data surface or invite every external agent into the transaction. It should discover a bounded communication surface and issue a minimally sufficient appeal.

The third requirement is semantic matching. External expertise may not be labeled using the originating institution's ontology. The system needs enough reasoning flexibility to recognize that a scientific-validation capability is relevant to investment risk even though it is not a finance function.

The fourth requirement is institutional interoperability. Financial information is often confidential, privileged, regulated, or market-sensitive. External experts may operate under different standards of disclosure and verification. A serious agentic system must reason about those differences rather than merely moving data between APIs.

The fifth requirement is authority partitioning. The expert who validates a technical claim should not automatically acquire the right to approve an acquisition. Conversely, the banker who owns the transaction should not automatically acquire the right to override scientific evidence. The compact should make decision rights explicit.

The sixth requirement is organizational learning. If a cross-domain collaboration reveals a superior verification practice, the institution should be able to evaluate whether that practice should become part of future workflow.

That is where the system moves from task automation to institutional improvement.

These ideas connect closely with the governance-first architecture in Reynoso's algorithmic-trading systems materials, where models, validation, risk, audit, and human review are treated as distinct layers rather than collapsed into a single autonomous trader [10]. The same principle applies here at a broader organizational scale: different functions can remain separately governed while still participating in a common decision ecology.

The implications extend beyond investment banking. In asset management, portfolio agents could seek external domain expertise when a thesis depends on science, regulation, or geopolitical structure. In risk management, market-risk agents could form temporary compacts with cyber or operational-risk societies when stress events cross traditional silos. In financial supervision, regulatory agents could interact with institution-specific agents under tightly defined information and authority boundaries. In insurance, underwriting societies could connect with climate, medical, engineering, or fraud societies when local competence becomes insufficient.

The common structure is the same: **local expertise remains primary; boundary crossing is selective; interoperability is explicit; learning can return home.**

This architecture also offers an alternative to the fashionable idea of one universal enterprise agent. A universal agent may be attractive because it simplifies interface design, but it concentrates context, authority, and failure modes. An ecology of specialized societies may instead preserve institutional separation while permitting intelligence to flow when justified. That design resembles the actual structure of sophisticated financial firms more closely than a single omniscient digital employee.

The notebook therefore invites students to think like system architects rather than merely model users. The key question is not "Which LLM should the bank deploy?" It is "What organizational architecture should surround multiple forms of intelligence so that competence can combine without authority becoming confused?"

The Research Frontier: From Societies to Agentic Economies

The notebook stops deliberately before one major frontier: economic incentives. As discussed earlier, the agents are mission-driven. They do not negotiate compensation. That choice isolates institutional emergence, but it also reveals the natural next question.

Once societies can escape, form cross-border coalitions, and import institutions, why would external agents participate? What if several societies want the same scarce specialist? What if one society contributes most of the value? What if an imported institutional rule produces benefits for years after the original coalition dissolves?

Introducing compensation would transform the ecology into something closer to an economy. Agents could receive direct contribution rewards, coordination rewards, risk compensation, or institutional royalties. Coalitions could have budgets. Societies could negotiate surplus sharing. Reputation could affect future access. The system would then need to solve credit assignment across individual, collective, and meta-constellation levels.

Another frontier concerns discovery. In NB05, the possibility of learning that a foreign board exists is supplied by the environment. A more endogenous system might learn which external communication surfaces are worth monitoring, develop reputational maps of societies, or create new public surfaces when repeated competence frontiers appear.

A third frontier concerns constitutional evolution. NB05 permits a local institutional mutation imported from another society. Future systems could allow constitutions to accumulate amendments, compare outcomes under different rules, or revert changes that create undesirable second-order effects. At that point institutional learning itself would become recursive.

A fourth frontier concerns conflict. The notebook studies cooperative boundary crossing. Real ecologies contain strategic behavior, incompatible objectives, deception, bargaining, and competition for resources. Multi-agent theory has long recognized that agent interaction can be cooperative or self-interested [1]. Introducing strategic incentives would substantially change the dynamics of trust and interoperability.

A fifth frontier concerns scale. Three societies make the mechanism visible. A real ecology might contain hundreds or thousands of specialized societies. Global discovery would then become a problem of search, reputation, routing, and institutional compatibility. It may be necessary to create higher-order registries, protocols, or market mechanisms without recreating the omniscient coordinator the architecture is trying to avoid.

A sixth frontier concerns evaluation. We need metrics not only for task success but for organizational quality. Did the escape reduce uncertainty? Did the compact preserve confidentiality? Did the coalition create value beyond local alternatives? Did the imported rule improve future decisions? Did the system become more robust or merely more connected?

The research program represented in the author's repositories on autonomous systems, agent architectures, papers, and course materials provides a natural home for such extensions [6, 7, 8, 9]. The key methodological principle should

remain the same: make mechanisms inspectable. Emergence should not become an excuse for mysticism.

The most exciting possibility is that future AI systems may develop increasingly sophisticated forms of organizational intelligence without requiring a single central mind. Problems could summon temporary organizations. Organizations could negotiate temporary constitutions. Successful practices could diffuse across society boundaries. Repeated interaction could produce reputational and institutional memory.

At that point the object of study would no longer be only artificial intelligence. It would be artificial institutional evolution.

Conclusion: From Autonomous Agents to Autonomous Ecologies

The journey of this course begins with a familiar object: the model. A model receives information and produces an answer. The first expansion is the agent, which can choose what to do next. The next expansion is the constellation, in which several agents coordinate around a problem. Adaptive organizations make the pattern of coordination variable. Endogenous coalitions make relationships contingent on the problem. Endogenous constitutions make collaboration rules partially subject to collective choice. Artificial societies accumulate local identity, memory, and institutional history.

NB05 asks what happens when the society itself is no longer treated as the final container.

The answer begins with local competence, not with global connectivity. Society A, B, and C are useful before they ever interact. Their constitutions shape how they organize internally. Their agents accumulate discoveries and relationships. Their local constellations solve problems. This matters because an ecology should not be a substitute for coherent local organization.

The turning point is the competence frontier. Society A recognizes that a technically decisive question lies outside its capabilities. The society understands the economic significance of the uncertainty but lacks engineering, patent, manufacturing, and scientific-validation expertise. The ability to represent that limit is itself a form of intelligence.

The first escape then changes the world. A local synthesis agent discovers a foreign communication surface and externalizes a bounded question. Information crosses. When a foreign specialist independently recognizes relevance, a relationship crosses. The two societies are now connected.

But connection immediately creates a governance problem. Finance and engineering do not collaborate under identical rules. Confidentiality and replication can conflict. Controlled sharing and peer criticism can conflict. Economic relevance and falsifiability represent different institutional priorities. The escape therefore produces an institutional collision.

The system does not solve that collision by centralizing. Instead, it creates a temporary compact. The compact defines what may be exchanged, how verification works, who may join, who decides, how provenance is preserved, when the relationship ends, and what may return home. A cross-society meta-constellation becomes possible without either society surrendering its identity.

The deepest event occurs on return. Information crosses back. A method crosses back. Finally, an institutional idea crosses back. Society A considers whether to adopt an adapted independent-verification practice. If it does, the society has learned organizationally.

That is the central message of the capstone: **self-improvement can occur at the level of organization**. The agents need not rewrite their own weights. The society can become better because its relationships, procedures, and institutions change in response to experience.

The final ecology therefore contains several forms of intelligence simultaneously. Individual agents reason. Local constellations coordinate. Societies govern. Meta-constellations bridge institutions. Compacts define temporary jurisdictions. Ledgers preserve memory. Institutional mutations encode lessons from prior interaction.

Autonomy has migrated upward.

At first, the answer becomes autonomous. Then the action. Then coordination. Then organization. Then relationships. Then coalitions. Then rules. Finally, the boundary itself becomes partially endogenous because the system can decide that the organization currently available is insufficient and that a governed bridge must be created beyond it.

That final step should not be confused with unrestricted emergence. The objective is not a system that silently invents organizations beyond human visibility. The objective is bounded emergence: enough openness for unexpected collaboration, enough structure for governance to remain possible.

For financial practitioners, this may be the more realistic path toward advanced autonomous AI. Banks, funds, advisers, exchanges, regulators, and research organizations are already societies of specialized intelligence. The future may not belong to one universal financial agent. It may belong to ecologies of bounded agentic institutions that know what they know, recognize what they do not know, reach outward selectively, negotiate interoperability, and

bring useful practices home.

The course therefore ends not with an answer but with a question.

If intelligent systems can increasingly determine what to do, whom to work with, how to organize, under what rules to collaborate, and when to cross the boundaries of their existing organizations, what exactly remains fixed in an autonomous system?

The problem helps produce the organization that solves the problem.

When the organization reaches the frontier of its own competence, intelligence may begin to reorganize not only what exists inside the boundary, but the boundary itself.

At that point, we are no longer studying only autonomous agents or even autonomous organizations. We are beginning to study the **ecology of autonomous systems**.

Notebook Architecture at a Glance

The notebook’s ten code units are best read as one continuous organizational experiment. The table below is intentionally compact; the body of the paper provides the full pedagogical interpretation.

Unit	Organizational move	Central question
1	Three local societies	Can we create independent artificial institutional worlds without exposing a global social map?
2	Local life	Can those worlds generate useful discoveries and relationships before external interaction is available?
3	Local constitutions	Do different rules create different collaboration patterns even when the basic agent architecture is similar?
4	Competence frontier	Can a society represent that an important problem exceeds its own capability set?
5	Informational escape	Can an agent externalize a bounded request without receiving a global directory of expertise?
6	Relational escape	Can foreign agents independently recognize semantic relevance and create a persistent cross-society relationship?
7	Institutional collision	What happens when technically complementary agents are governed by incompatible local rules?
8	Cross-society compact	Can a temporary bridge institution enable collaboration without replacing either constitution?
9	Institutional mutation	Can information, methods, and institutional practices return home and change future organization?
10	Ecology	Can several societies remain locally autonomous while forming selective, governed interdependencies?

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Source note. This image-free version of the paper is designed to accompany the Colab notebook `TOPIC_772_ESCAPES.ipynb`. The infographic pages are intentionally omitted.